

SVD of gradient

$$\mathbf{G} = \mathbf{U} \cdot \mathbf{\Sigma} \cdot \mathbf{V}^\top$$

Gradient

$\sigma_1, \sigma_2, \dots$

Muon
 $\Sigma \rightarrow \mathbf{I}$

$$\mathbf{U} \cdot \mathbf{I} \cdot \mathbf{V}^\top$$

All $\sigma_i = 1$

$= \mathbf{U}\mathbf{V}^\top$ (orthogonal update)

Muon update